SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – APPLICATION BASED QUESTIONS

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO2	**Assessment:**	Application Based Questions, Inclass
Weightage:	_____ 40% of CIA	**Duration / Marks:**	60 mins
CO2 Statement:	Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)		
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Section / Batch:	D/2026	**Date:**	21/10/2026

Code of Conduct

I G.SANGHAMITRA (1924260351) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each question carefully before answering (in-Class Assessment)
- Each question is a real-world application scenario. Identify the AI technique that fits, apply it step by step, and justify every decision.
- Questions are drawn from Informed Search (Greedy, A*, heuristics), Local Search and Online Search, Constraint Satisfaction Problems, and Adversarial Search (Minimax, Alpha-Beta, evaluation functions).
- Show intermediate tables (open lists, domains, α/β values) and state the final answer clearly.
- Marks will be awarded using the Application Based Questions rubric: understanding of problem context, application of concepts, problem-solving approach, accuracy of solution and clarity.
- Upload your work in GCR and Github.

Section / Topic	Focus	Question Nos.
1. Informed Search Strategies	Greedy best-first search, A* search, heuristic functions (admissibility, consistency)	**Q1**
2. Local Search, Online Search & Constraint Satisfaction	Local search and optimization problems, online search agent, CSP (MRV, degree heuristic, forward checking)	**Q2, Q3**

3. Game Playing: Adversarial Search	Games, minimax algorithm, alphabeta pruning, evaluation functions	**Q4**

Rubrics for Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Understanding of Problem Context	**20%**	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	**30%**	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	**20%**	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	**20%**	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	**10%**	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach (2)	Accuracy of Solution (2)	Clarity (1)	Sub. Total (10)	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____

Date: _____	Date: _____	Date: _____

Annexure A – Application Based Questions

Question 1: Campus Delivery Robot – Greedy Best-First vs A*

Topic: Informed search strategies, heuristic functions | 15 minutes

A campus delivery robot must carry a parcel from the Store (S) to the Hostel (G). The one-way road network and the robot's heuristic estimate h(n) of the remaining travel cost to G (in minutes) are given below.

From	To	Cost
S	A	1
S	B	4
A	C	2
A	D	5
B	G	5
C	D	1
D	G	2

Node	S	A	B	C	D	G
h(n)	6	5	3	3	2	0

Assume ties are broken alphabetically and that a node is re-opened if a cheaper path to it is found.

(a) Apply Greedy Best-First Search from S to G. Write the order in which nodes are expanded, the path returned and its total cost.

(b) Apply A* search. Show the open list with g(n), h(n) and f(n) after each expansion, the order of expansion, and the final path and cost.

(c) Check whether h(n) is admissible and consistent by verifying the required condition for every node and every road.

(d) Compare the outputs of (a) and (b). Explain why Greedy Best-First Search returned a different result and which algorithm you would deploy for the robot, with justification.

Question 2: Wi-Fi Access Point Placement – Local Search and Online Search

Topic: Local search algorithms and optimization problems, online search agent | 15 minutes

A technician wants to place a Wi-Fi access point at the best position along a corridor with 10 candidate positions x = 0 to 9. The measured signal quality f(x) (higher is better) is:

x	0	1	2	3	4	5	6	7	8	9
f(x)	2	5	9	6	4	7	12	10	8	3

A move shifts the access point to an adjacent position ($x - 1$ or $x + 1$). Steepest-ascent hill climbing moves to the better neighbour and stops when no neighbour is strictly better.

(a) Run steepest-ascent hill climbing from the start positions x = 0, x = 4 and x = 9. Show each step and the final position and signal quality.

(b) Identify all local maxima and the global maximum. If the start position is chosen uniformly at random from the 10 positions, find the probability that one run reaches the global maximum, the probability that at least one of 3 random-restart runs does, and the expected number of runs needed.

(c) A simulated annealing search is at x = 2 and proposes the move to x = 3. Compute the acceptance probability e^(Δ/T) at T = 2 and at T = 1, and explain what this does to the chance of escaping the local maximum as T falls.

(d) The same search is later done by a robot that cannot see the whole corridor, can only measure f(x) where it stands, and cannot undo exploration for free. Explain why this is an online search problem, and state what the agent must remember to avoid cycling.

Question 3: Exam Timetable Scheduling – Constraint Satisfaction Problem

Topic: Constraint satisfaction problem (MRV, degree heuristic, forward checking) | 15 minutes

Five exams A, B, C, D and E must each be given one slot from {1, 2, 3}. Two exams that share students must not be in the same slot. The pairs sharing students are: A–B, A–C, A–D, B–C, B–D, C–E, D–E. Because of invigilator availability, exam A can only be in slots 1 or 2, and exam E only in slots 1 or 2.

Exam	Initial domain
A	{1, 2}
B	{1, 2, 3}

C	{1, 2, 3}
D	{1, 2, 3}
E	{1, 2}

Rules for the search: choose the variable using Minimum Remaining Values (MRV), break ties with the degree heuristic, then alphabetically; try values in ascending order; use forward checking after every assignment.

(a) Draw the constraint graph (list its edges) and compute the degree of every exam.

(b) Select the first exam to assign using MRV and the degree heuristic, justify the choice, assign its first value, and show the domains of all exams after forward checking.

(c) Continue the search with the same rules, showing the chosen exam, its value and the updated domains at each step, until a complete assignment is found. Verify it against every constraint.

(d) Interpret: explain how MRV, the degree heuristic and forward checking reduced the search effort here, and what forward checking would signal if a domain became empty.

Question 4: Board-Game AI – Minimax, Alpha-Beta Pruning and Evaluation Function

Topic: Game playing, adversarial search, evaluation functions | 15 minutes

A game-playing agent (MAX) searches two plies ahead: MAX chooses one of three moves (B, C, D), then MIN replies with one of three moves. Positions at depth 2 are scored by the evaluation function

$$\mathbf{Eval = 2 \times M + 1 \times Mo}$$

where M is the material advantage of MAX and Mo is the mobility advantage of MAX. The leaf features (M, Mo) are given for the leaves in left-to-right order:

MIN node	Leaf 1 (M, Mo)	Leaf 2 (M, Mo)	Leaf 3 (M, Mo)
B	(1, 1)	(5, 2)	(3, 2)
C	(1, 0)	(1, 2)	(2, 2)
D	(6, 2)	(2, 1)	(0, 2)

(a) Compute the Eval value of all nine leaves.

(b) Run plain minimax. Give the backed-up value of B, C and D, the value at the root and the move MAX should play.

(c) Run alpha-beta pruning on the same tree (left to right). Show α and β at each node, mark every pruned branch, and state the root value and the number of leaves evaluated.

(d) If the leaves under C were ordered (6, 4, 2) instead, how many leaves would alpha-beta evaluate? Explain the effect of move ordering and why pruning never changes the minimax value.

Q1. Campus Delivery Robot – Greedy Best-First vs A*

(a) Apply Greedy Best-First Search. Show expansion order, returned path, and total cost.

Answer:

Greedy Best-First Search selects the node with the lowest heuristic value $h(n)$. It does not consider the actual path cost when selecting a node.

Given heuristic values:

Node	Heuristic $h(n)$
S	6
A	5
B	3
C	3
D	2
G	0

Step 1: Start at S.

- Expand S.
- Its successors are A and B.
- A has $h(A)=5$, and B has $h(B)=3$.
- Select B because it has the lower heuristic.

Step 2: Expand B.

- B leads to G.
- $h(G)=0$, so select G.

Expansion order: S → B → G

Returned path: S → B → G

Total cost:

$$ S\rightarrow B\rightarrow G=4+5=\boxed{9} $$

Final answer: Greedy Best-First Search returns the path S → B → G with a total cost of 9 minutes.

(b) Apply A* search. Show the OPEN list after each expansion, expansion order, final path, and cost.

Answer:

A* uses the evaluation function:

$$ f(n)=g(n)+h(n) $$

Here, $g(n)$ is the actual cost from S to node n, and $h(n)$ is the estimated remaining cost.

The node with the smallest $f(n)$ is expanded first.

For clarity, OPEN lists below are sorted by increasing $f(n)$.

Step 1: Start at S

$$ g(S)=0,\quad h(S)=6,\quad f(S)=6 $$

Expand S and generate A and B.

- A: $g=1,\ h=5,\ f=6$
- B: $g=4,\ h=3,\ f=7$

OPEN node	$g(n)$	$h(n)$	$f(n)$
A	1	5	6
B	4	3	7

Step 2: Expand A

A has the smallest $f(n)=6$.

Generate C and D.

- C: $g=1+2=3,\ h=3,\ f=6$
- D: $g=1+5=6,\ h=2,\ f=8$

OPEN node	$g(n)$	$h(n)$	$f(n)$
C	3	3	6
B	4	3	7
D	6	2	8

Step 3: Expand C

C has the smallest $f(n)=6$.

C leads to D with a new path cost:

$$ g(D)=3+1=4 $$

This is cheaper than the previous cost of 6, so update D.

$$ f(D)=4+2=6 $$

OPEN node	$g(n)$	$h(n)$	$f(n)$
D	4	2	6
B	4	3	7

Step 4: Expand D

D leads to G.

$$ g(G)=4+2=6 $$

$$ f(G)=6+0=6 $$

OPEN node	$g(n)$	$h(n)$	$f(n)$
G	6	0	6
B	4	3	7

Step 5: Expand G

G is the goal, so the search terminates.

Expansion order: S → A → C → D → G

Final path:

$$ S\rightarrow A\rightarrow C\rightarrow D\rightarrow G $$

Total cost:

$$ 1+2+1+2=\boxed{6\text{ minutes}} $$

Final answer: A* returns the path S → A → C → D → G with a total cost of 6 minutes.

(c) Check whether the heuristic is admissible and consistent.

Answer:

A heuristic is admissible if it never overestimates the actual minimum cost to reach the goal.

$$ h(n)\leq h^*(n) $$

Here, $h^*(n)$ is the true minimum remaining cost.

The shortest costs to G are:

Node	Given $h(n)$	Actual shortest cost $h^*(n)$	Admissible?
S	6	6	Yes
A	5	5	Yes
B	3	5	Yes

C	3	3	Yes
D	2	2	Yes
G	0	0	Yes

Therefore, the heuristic is admissible because it never overestimates the true minimum cost.

A heuristic is consistent if every directed edge satisfies:

\[h(u)\leq c(u,v)+h(v) \]

Check every road:

Road	Verification	Result
S → A	\(6\leq1+5=6\)	True
S → B	\(6\leq4+3=7\)	True
A → C	\(5\leq2+3=5\)	True
A → D	\(5\leq5+2=7\)	True
B → G	\(3\leq5+0=5\)	True
C → D	\(3\leq1+2=3\)	True
D → G	\(2\leq2+0=2\)	True

Final answer: The heuristic is both admissible and consistent, since all the required conditions are satisfied.

(d) Compare Greedy Best-First Search and A*. Recommend an algorithm.

Answer:

Feature	Greedy Best-First Search	A* Search

Evaluation	$f(n)=h(n)$	$f(n)=g(n)+h(n)$
Main consideration	Estimated remaining cost	Actual cost plus estimated remaining cost
Returned path	S → B → G	S → A → C → D → G
Total cost	9 minutes	6 minutes
Optimality	Not guaranteed	Guaranteed under suitable conditions

Greedy Best-First Search selects B because its heuristic value is lower than A's. However, the actual cost of travelling through B and G is 9 minutes.

A* considers both actual and estimated costs. It discovers a cheaper route through A, C, and D, with a total cost of 6 minutes.

Final recommendation: A* is the better choice for the campus delivery robot because it finds the minimum-cost route under the given conditions. The heuristic is admissible and consistent, so A* can guarantee an optimal solution for this graph.

Q2. Wi-Fi Access Point Placement – Local Search and Online Search

(a) Run steepest-ascent Hill Climbing from x = 0, x = 4, and x = 9.

Answer:

Steepest-ascent Hill Climbing moves to the adjacent position with the highest signal quality, provided that it is strictly better than the current position.

Given signal quality:

Position x	0	1	2	3	4	5	6	7	8	9
$f(x)$	2	5	9	6	4	7	12	10	8	3

Case 1: Starting at $x=0$

- $x=0,\ f(0)=2$
- Move to $x=1,\ f(1)=5$.

- Move to $x=2,\ f(2)=9$.
- The neighbouring values are $f(1)=5$ and $f(3)=6$. Neither is better than 9.

Path: $0\rightarrow1\rightarrow2$

Final position: $\boxed{x=2}$

Final signal quality: $\boxed{9}$

Case 2: Starting at $x=4$

- $x=4,\ f(4)=4$
- Neighbours have values 6 and 7, so move to $x=5$.
- Move to $x=6,\ f(6)=12$.
- Neither neighbour of $x=6$ has a higher value.

Path: $4\rightarrow5\rightarrow6$

Final position: $\boxed{x=6}$

Final signal quality: $\boxed{12}$

Case 3: Starting at $x=9$

- $x=9,\ f(9)=3$
- Move to $x=8,\ f(8)=8$.
- Move to $x=7,\ f(7)=10$.
- Move to $x=6,\ f(6)=12$.
- No neighbouring position has a higher value.

Path: $9\rightarrow8\rightarrow7\rightarrow6$

Final position: $\boxed{x=6}$

Final signal quality: $\boxed{12}$

Conclusion: Hill Climbing may reach different final positions depending on its starting point.

(b) Identify all local maxima and the global maximum. Calculate the probabilities.

Answer:

A local maximum is a position whose signal quality is higher than that of its neighbouring positions. The global maximum is the position with the highest signal quality across the entire corridor.

Local maxima:

- At $x=2$, $f(2)=9$. Its neighbours have values 5 and 6.
- At $x=6$, $f(6)=12$. Its neighbours have values 7 and 10.

Therefore:

- Local maxima: $\boxed{x=2,\ x=6}$
- Global maximum: $\boxed{x=6,\ f(6)=12}$

1. Probability that one random start reaches the global maximum

Starting positions that reach $x=6$ are $4,5,6,7,8,9$.

Number of successful starting positions = 6.

Total starting positions = 10.

$$P=\frac{6}{10}=\boxed{0.6=60\%}$$

2. Probability that at least one of three independent random-restart runs reaches the global maximum

The probability that one run fails is:

$$1-0.6=0.4$$

The probability that all three runs fail is:

$$(0.4)^3=0.064$$

Therefore:

$$ P(\text{at least one success})=1-0.064 $$

$$ \boxed{0.936=93.6\%} $$

3. Expected number of runs needed

For independent runs with success probability $p=0.6$, the expected number of runs until the first success follows a geometric distribution.

$$ E(N)=\frac{1}{p} =\frac{1}{0.6} =\boxed{1.67\text{ runs (approximately)}} $$

Final answers:

- Probability for one run: 60%
- Probability for at least one success in three runs: 93.6%
- Expected runs until success: 1.67 runs

(c) Simulated Annealing: Calculate the acceptance probability at T = 2 and T = 1.

Answer:

The search is at $x=2$ and proposes moving to $x=3$.

Given:

$$ f(2)=9,\qquad f(3)=6 $$

The change in signal quality is:

$$ \Delta=f(3)-f(2)=6-9=-3 $$

For a worse move, Simulated Annealing accepts the move with probability:

$$ P=e^{\Delta/T} $$

At $T=2$:

$$ P=e^{-3/2}=e^{-1.5} $$

$$\boxed{P\approx0.2231=22.31\%}$$

At $T=1$:

$$P=e^{-3/1}=e^{-3}$$

$$\boxed{P\approx0.0498=4.98\%}$$

Explanation:

At a higher temperature, the algorithm is more likely to accept a worse move. This helps it escape local maxima and explore other parts of the search space. As the temperature decreases, the acceptance probability for worse moves decreases.

Final answer: The acceptance probability decreases from approximately 22.31% at $T=2$ to 4.98% at $T=1$. Thus, the algorithm becomes less likely to accept worse moves as the temperature falls.

(d) Explain why the robot's problem requires online search and what it must remember to avoid cycling.

Answer:

This is an online search problem because the robot cannot observe the entire corridor in advance. It can measure signal quality only at its current position and must make decisions while exploring.

The robot faces the following challenges:

- It has incomplete information about the corridor.
- It discovers signal quality gradually.
- It cannot undo previous exploration without cost.
- It must select its next action using available observations.

To avoid cycling, the robot should remember:

1. Previously visited positions.
2. Signal quality measured at each visited position.
3. Actions or transitions already attempted.

4. Its current position and useful information about neighbouring positions.

This information helps the robot avoid repeatedly visiting the same positions without making progress.

Conclusion: An online search agent with memory of visited states and previous actions is suitable for the robot because it can learn about the environment during exploration.

Q3. Exam Timetable Scheduling – Constraint Satisfaction Problem (CSP)

(a) Draw the constraint graph and calculate the degree of every exam.

Answer:

A Constraint Satisfaction Problem (CSP) consists of variables, domains, and constraints. Here, each exam is a variable, and each exam must be assigned one of the available slots \(\{1,2,3\}\), subject to the given restrictions.

The constraint graph contains five vertices: A, B, C, D, and E.

Edges (exams that share students):

\[\{AB, AC, AD, BC, BD, CE, DE\} \]

The degree of a vertex is the number of edges connected to it.

Exam	Connected exams	Degree
A	B, C, D	3
B	A, C, D	3
C	A, B, E	3
D	A, B, E	3
E	C, D	2

Final answer: A, B, C, and D have degree 3, while E has degree 2.

(b) Select the first exam using MRV and the degree heuristic. Assign its first value and show the updated domains.

Answer:

Minimum Remaining Values (MRV) selects the variable with the smallest domain. If multiple variables have the same domain size, the degree heuristic selects the variable connected to the greatest number of unassigned variables. If there is still a tie, alphabetical order is used.

Initial domains:

Exam	Domain	Domain size
A	12	2
B	123	3
C	123	3
D	123	3
E	12	2

Step 1: Select a variable

MRV selects A or E because both have two values. The degree heuristic selects A because A has degree 3, while E has degree 2.

Therefore, select A.

Step 2: Assign the first value

Values are tried in ascending order, so assign:

\[A=1 \]

Step 3: Apply forward checking

A shares students with B, C, and D. Therefore, none of those exams can use slot 1.

Updated domains:

Exam	Domain after forward checking
A	1
B	23
C	23
D	23
E	12

Final answer: The first exam selected is A, and its assigned slot is 1.

(c) Continue the search until a complete assignment is found. Show every step and verify the constraints.

Answer:

Continue using MRV, the degree heuristic, alphabetical tie-breaking, ascending value order, and forward checking.

Step 1: A = 1

After forward checking:

- A = 1
- B = 3
- C = 3
- D = 3
- E = 2

Step 2: Select B

B, C, and D have the smallest remaining domain size of 2 among unassigned variables. Their degrees are all 3.

Alphabetical order selects B.

Assign the smallest available value:

$$ B=2 $$

Forward checking removes slot 2 from C and D because both share students with B.

Updated domains:

Exam	Remaining domain
A	1
B	2
C	3
D	3
E	12

Step 3: Select C

C and D each have one remaining value. Their degrees are equal, so alphabetical order selects C.

Assign:

$$ C=3 $$

Forward checking does not remove any additional values from E because E's domain is 2, and C is assigned slot 3.

Updated domains:

Exam	Remaining domain
A	1
B	2
C	3

D	3
E	12

Step 4: Select D

D has only one available value.

Assign:

$$ D=3 $$

D and E share students, so slot 3 is excluded from E's domain. Since E's domain was already 2, it remains unchanged.

Updated domains:

Exam	Remaining domain
A	1
B	2
C	3
D	3
E	12

Step 5: Select E

E is the only unassigned variable. Try the smallest available value:

$$ E=1 $$

All exams have now been assigned.

Final timetable

Exam	Assigned slot

A	1
B	2
C	3
D	3
E	1

Verification of all constraints:

Constraint	Verification	Status
A–B	$1 \neq 2$	Satisfied
A–C	$1 \neq 3$	Satisfied
A–D	$1 \neq 3$	Satisfied
B–C	$2 \neq 3$	Satisfied
B–D	$2 \neq 3$	Satisfied
C–E	$3 \neq 1$	Satisfied
D–E	$3 \neq 1$	Satisfied

Final answer: The valid assignment is A = 1, B = 2, C = 3, D = 3, E = 1. Every given constraint is satisfied.

(d) Explain how MRV, degree heuristic, and forward checking reduce search effort.

Answer:

1. Minimum Remaining Values (MRV)

MRV selects the exam with the fewest available slots. This helps identify difficult assignments early and reduces unnecessary exploration.

2. Degree heuristic

When MRV produces a tie, the degree heuristic selects the exam that constrains the largest number of other unassigned exams. This helps resolve highly connected variables earlier.

3. Forward checking

Forward checking removes invalid values from the domains of neighbouring exams after each assignment. This detects potential conflicts before the entire timetable is completed.

If a domain becomes empty:

If an unassigned exam has no valid slot remaining, the current assignment cannot lead to a valid solution. The algorithm must backtrack and try another value for an earlier variable.

Conclusion: Combining MRV, the degree heuristic, and forward checking reduces the search space and helps the CSP solver find a valid timetable efficiently.

Q4. Board-Game AI – Minimax, Alpha-Beta Pruning and Evaluation Function

(a) Compute the evaluation value of all nine leaves.

Answer:

The evaluation function is:

\[Eval=2M+Mo \]

Where:

- \(M\) = material advantage of MAX.
- \(Mo\) = mobility advantage of MAX.

Substitute the given values for each leaf.

MIN node	Leaf	\(M\)	\(Mo\)	Calculation	Eval
B	1	1	1	\(2(1)+1\)	3

B	2	5	2	\(2(5)+2\)	12
B	3	3	2	\(2(3)+2\)	8
C	1	1	0	\(2(1)+0\)	2
C	2	1	2	\(2(1)+2\)	4
C	3	2	2	\(2(2)+2\)	6
D	1	6	2	\(2(6)+2\)	14
D	2	2	1	\(2(2)+1\)	5
D	3	0	2	\(2(0)+2\)	2

Final answer: The nine leaf evaluation values are:

- B: 3, 12, 8
- C: 2, 4, 6
- D: 14, 5, 2

(b) Run Minimax. Find the backed-up values of B, C, and D, the root value, and the best move.

Answer:

Minimax assumes that MAX chooses the highest value and MIN chooses the lowest value.

Step 1: Calculate the value of B

B is a MIN node.

\[B=\min(3,12,8)=\boxed{3} \]

Step 2: Calculate the value of C

C is a MIN node.

\[C=\min(2,4,6)=\boxed{2} \]

Step 3: Calculate the value of D

D is a MIN node.

$$ D=\min(14,5,2)=\boxed{2} $$

Step 4: Calculate the root value

The root is a MAX node.

$$ Root=\max(B,C,D) $$

$$ =\max(3,2,2)=\boxed{3} $$

Move	MIN's backed-up value
B	3
C	2
D	2

Final answer: The root value is 3, and MAX should choose move B.

(c) Run Alpha-Beta pruning from left to right. Show alpha, beta, pruned branches, root value, and number of leaves evaluated.

Answer:

Alpha-Beta pruning improves Minimax by eliminating branches that cannot affect the final decision.

- α: Best value MAX can guarantee so far.
- β: Best value MIN can guarantee so far.
- Prune when $\alpha\geq\beta$.

Initially:

$$ \alpha=-\infty,\qquad\beta=+\infty $$

Step 1: Explore B

B is a MIN node. Evaluate its leaves from left to right.

Leaf	Value	Beta at B	Action
B1	3	3	Continue
B2	12	3	Continue
B3	8	3	Finish B

Therefore:

$$ B=\min(3,12,8)=3 $$

At the root, update:

$$ \alpha=\max(-\infty,3)=3 $$

Three leaves have been evaluated.

Step 2: Explore C

C is a MIN node, entered with $\alpha=3$.

- Evaluate C1 = 2.
- Update $\beta=\min(+\infty,2)=2$.
- Now $\alpha=3$ and $\beta=2$.

Since:

$$ \alpha\geq\beta \quad (3\geq2) $$

prune the remaining branches C2 and C3.

Therefore:

$$ C=2 $$

Only one leaf under C is evaluated.

Step 3: Explore D

D is a MIN node, entered with $\alpha=3$.

Leaf	Value	Beta at D	Pruning?
D1	14	14	No
D2	5	5	No
D3	2	2	No remaining leaves

At D1, $\beta=14$, so $3\geq14$ is false.

At D2, $\beta=5$, so $3\geq5$ is false.

At D3, $\beta=2$, so $3\geq2$ is true, but there are no remaining branches under D to prune.

Thus:

$$ D=\min(14,5,2)=2 $$

Final backed-up values:

Node	Value
B	3
C	2
D	2
Root (MAX)	3

Pruned branches: C2 and C3.

Number of leaves evaluated:

$$ 3+1+3=\boxed{7} $$

Final answer: Alpha-Beta pruning evaluates 7 out of 9 leaves, prunes C2 and C3, and returns the same root value of 3. MAX should choose B.

(d) If the leaves under C are ordered (6, 4, 2), how many leaves will Alpha-Beta evaluate? Explain move ordering.

Answer:

After evaluating B, the root has:

$$\alpha=3$$

Now C's leaves are visited in the order 6, 4, 2.

Leaf under C	Value	Beta at C	Pruning condition
C1	6	6	$3\geq6$: False
C2	4	4	$3\geq4$: False
C3	2	2	$3\geq2$: True

All three leaves under C must be evaluated because the pruning condition is not met until the final leaf.

Under B, three leaves are evaluated. Under C, three leaves are evaluated. Under D, three leaves are evaluated.

$$\text{Total}=3+3+3=\boxed{9\text{ leaves}}$$

Effect of move ordering:

- With the original order (2, 4, 6), C's first leaf immediately causes pruning, so only 7 leaves are evaluated overall.
- With the order (6, 4, 2), pruning occurs only after all three C leaves have been evaluated, so 9 leaves are evaluated.

Alpha-Beta pruning never changes the minimax value because it removes only branches that cannot influence the final decision, provided the bounds and search are applied correctly.